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SOC

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MITM (Man-in-the-Middle Attack)

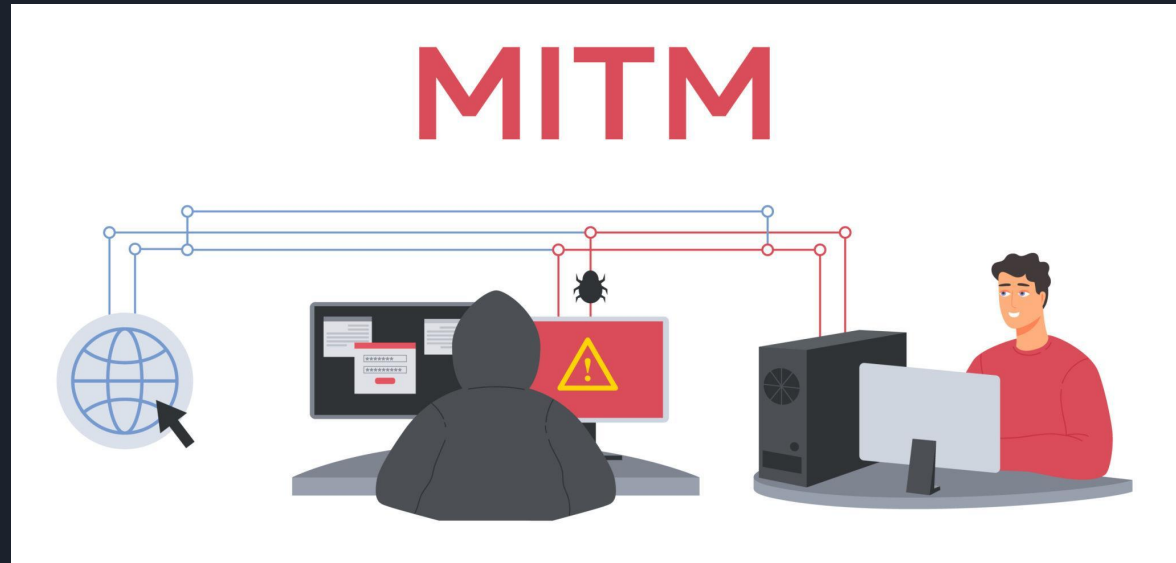
What is MITM?

A **Man-in-the-Middle (MITM)** attack happens when an attacker secretly intercepts communication between two parties.

- User $\rightleftharpoons$ Attacker $\rightleftharpoons$ Server

Instead of:

- User $\rightleftharpoons$ Server





MITM (Man-in-the-Middle Attack)

Goal of MITM

- Steal credentials
- Capture session cookies
- Modify data in transit
- Inject malware
- Monitor sensitive communication



MITM (Man-in-the-Middle Attack)

Types of MITM Attacks

ARP Spoofing (Very Important for SOC)

Attacker tricks devices into thinking:

"I am the router"

So traffic goes through attacker.

Example Flow :

Victim → Attacker → Router → Internet

SOC Indicators

- Duplicate MAC addresses
- ARP table changes
- Sudden traffic rerouting
- Internal network anomalies



MITM (Man-in-the-Middle Attack)

Types of MITM Attacks

DNS Spoofing

Attacker redirects domain to fake IP.

Example:

google.com → attacker server

Very similar to **pharming**.

SSL Stripping

Attacker downgrades:

HTTPS → HTTP

User thinks connection is secure but it's not.



MITM (Man-in-the-Middle Attack)

Types of MITM Attacks

Session Hijacking

Attacker steals session cookies and logs in without password.

Example:

- User logs into website
- Attacker steals cookie
- Attacker gains access

Evil Twin

Fake WiFi network:

- User connects
- Attacker intercepts all traffic



MITM (Man-in-the-Middle Attack)

Real SOC Scenario (MITM)

Alert:

- Multiple users accessing same gateway IP
- SSL certificate mismatch
- Suspicious DNS responses

Investigation:

- Check ARP table
- Check DNS logs
- Check network traffic anomalies

MITM Prevention

- HTTPS (TLS encryption)
- VPN usage
- Secure DNS (DNSSEC)
- Certificate validation
- Network segmentation
- WPA3 WiFi security



MITM (Man-in-the-Middle Attack)

DDoS (Distributed Denial of Service)

What is DDoS?

A DDoS attack floods a system with **massive traffic** to make it:

- Slow
- Unresponsive
- Completely down





MITM (Man-in-the-Middle Attack)

DDoS (Distributed Denial of Service)

Real SOC Example

Goal of DDoS :

- Take website offline
- Disrupt business operations
- Extortion (“Pay or we keep attacking”)
- Diversion for other attacks

Alert:

Traffic increased from 2k req/sec → 200k req/sec

Source: Multiple countries

Endpoint: /login

Likely: Application Layer DDoS



MITM (Man-in-the-Middle Attack)

Types of DDoS Attacks

Volumetric Attacks

Flood network bandwidth.

Example:

- UDP Flood
- ICMP Flood

Protocol Attacks

Target server resources.

Example:

- SYN Flood
- Ping of Death



MITM (Man-in-the-Middle Attack)

Types of DDoS Attacks

Application Layer Attacks

Target specific applications.

Example:

- HTTP Flood
- Login endpoint abuse

Harder to detect.

Real SOC Indicators

- Sudden spike in traffic
- Same requests from multiple IPs
- High CPU/memory usage
- Increased latency
- Service downtime



MITM (Man-in-the-Middle Attack)

DDoS Mitigation

- Rate limiting
- Web Application Firewall (WAF)
- CDN (Cloudflare, Akamai)
- Traffic filtering
- Load balancing
- IP blocking



MITM (Man-in-the-Middle Attack)

MITM vs DDoS (Important for Interview)

Feature	MITM	DDoS
Goal	Intercept data	Disrupt service
Impact	Data theft	Service outage
Visibility	Stealthy	Noisy
Detection	Harder	Easier
Type	Confidentiality attack	Availability attack



MITM (Man-in-the-Middle Attack)

SOC Analyst Thinking

If you see:

- Suspicious certificate warning
- Users complaining about login hijack
- DNS mismatch

Think it as **MITM**

If you see:

- Website down
- Traffic spike
- Server overload

Think it as **DDoS**